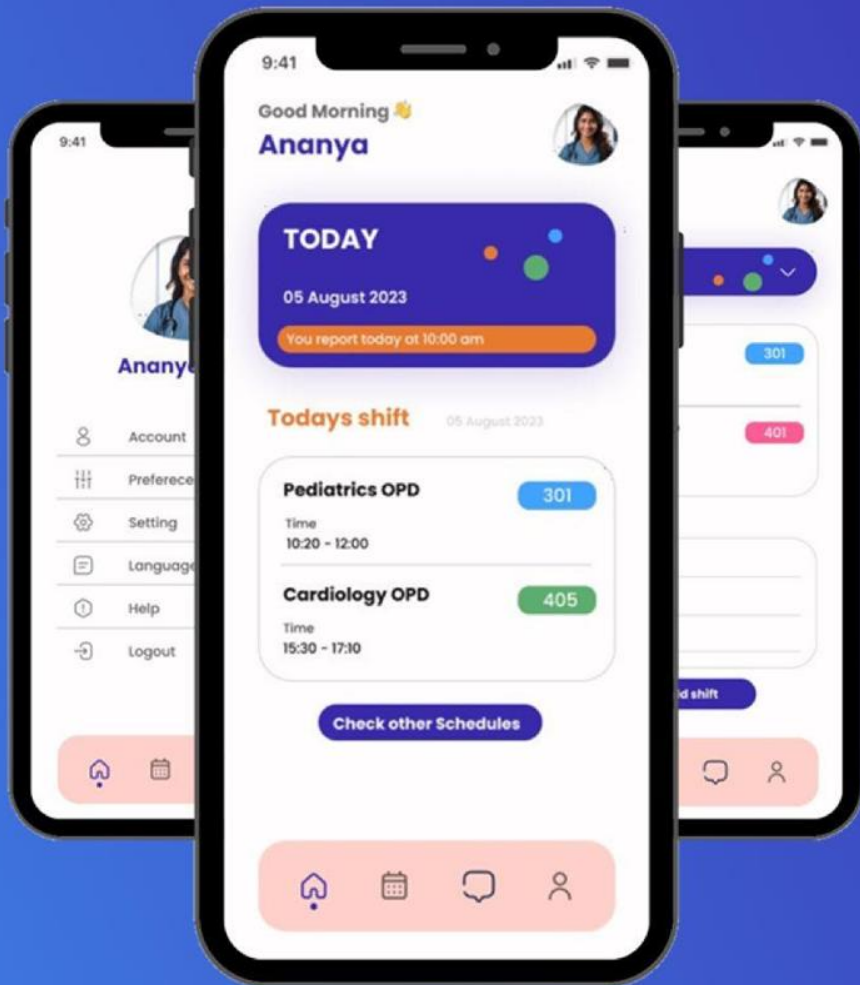




| ShiftCare

Hospital Staff Scheduling App

Ankita Thatte



Project overview



The product:

The hospital staff scheduling app streamlines scheduling for healthcare professionals, improving efficiency and satisfaction. Key features include an intuitive interface for shift management, seamless shift swapping, real-time notifications, personal calendar integration, compliance tracking, and robust reporting. Designed with a user-centric approach, it meets the specific needs of doctors, nurses, and administrators.



Project duration:

2 weeks

Project overview



The problem:

The current hospital staff scheduling processes are inefficient and cumbersome, leading to operational disruptions and dissatisfaction among healthcare professionals. This app aims to streamline scheduling, improve work-life balance, and enhance overall staff satisfaction.



The goal:

The goal is to create a user-friendly and efficient hospital staff scheduling app that streamlines the scheduling process, enhances operational efficiency, and improves work-life balance and satisfaction among healthcare professionals.

Project overview



My role:

UX Designer



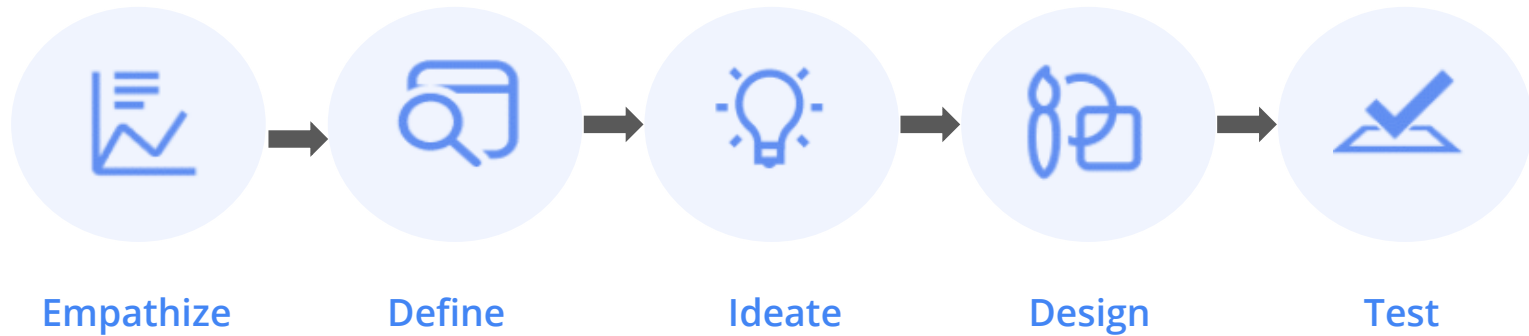
Responsibilities:

In this project, I focused on understanding the users' needs through research, creating intuitive designs, and developing prototypes. I also handled usability testing and worked closely with the development team to ensure the design met both user needs and technical constraints.

Understanding the user

- User research
- Personas
- Problem statements
- User journey maps

Design Thinking Process



User research: summary



In the user research phase, I interviewed hospital staff to understand their challenges with the current scheduling system. Key findings included difficulties in managing shift changes and a lack of real-time communication. These insights highlighted the need for a more flexible and intuitive scheduling tool that facilitates easier shift management and better communication.

User research: pain points

1

Complex Shift Changes

Staff experienced difficulties in making and managing shift changes due to a rigid scheduling system that lacked flexibility and real-time updates.

2

Inefficient Communication

Staff faced challenges in coordinating schedules and communicating with colleagues, leading to frequent misunderstandings and misalignment.

3

Limited access

Staff had trouble accessing scheduling information when away from work, leading to delays and difficulties in managing shifts effectively.

4

Error-prone Manual processes

Manual scheduling processes led to frequent errors and increased administrative work, impacting overall staff efficiency and satisfaction.

Persona: Dr. Priya Sharma

Problem statement:

Dr. Priya Sharma struggles with a time-consuming and inflexible scheduling system that disrupts her planning and coordination with her team, affecting her ability to manage her busy clinic and surgery hours efficiently.



Dr. Priya Sharma

Age: 38

Education: MBBS, MD

Hometown: Pune

Family: Husband, two children

Occupation: Senior Cardiologist

“Juggling a busy cardiology practice and family life, I need a scheduling system that helps me manage my shifts smoothly and adapt quickly to any last-minute changes.”

Goals

- Manage her complex schedule efficiently to balance work and family.
- Handle shift changes quickly with minimal disruption.

Frustrations

- Time-consuming scheduling systems that complicate managing her busy schedule.
- Frequent last-minute changes that disrupt her well-planned day and add stress.

Dr. Priya Sharma, a 38-year-old senior cardiologist, balances a demanding job with family responsibilities. She struggles with time-consuming scheduling systems and frequent last-minute changes that disrupt her well-organized life.

Persona: Ananya Patel

Problem statement:

Ananya Patel faces challenges with inconsistent schedules and poor communication about changes, making it difficult to manage her work-life balance and leading to stress and burnout.



Ananya Patel

Age: 29

Education: B.Sc Nursing

Hometown: Pune

Family: Divorced, 1 daughter

Occupation: Nurse

“Balancing long shifts as a single mother is tough. I need a scheduling system that makes it easier to manage my time with clear shifts, easy swaps, and timely updates.”

Goals

- Achieve a consistent work schedule to better balance time with her daughter and reduce stress.
- Enable easy shift swaps and timely notifications to handle unexpected changes and maintain a manageable routine.

Frustrations

- Inconsistent schedules that make it difficult to plan personal activities and manage time with her daughter.
- Lack of communication about schedule changes leading to last-minute adjustments and increased stress.

Ananya Patel is a 29-year-old pediatric nurse and single mother who struggles with inconsistent schedules and poor communication about shift changes. Her demanding job and family responsibilities make it challenging to balance her work and personal life.

Persona: **Rajesh Verma**

Problem statement:

Rajesh Verma deals with labor-intensive and error-prone manual scheduling, making it difficult to track staff availability, ensure optimal coverage, and maintain compliance with labor regulations and hospital policies.



Rajesh Verma

Age: 45

Education: MBA Healthcare

Hometown: Pune

Family: Wife, two children

Occupation: Hospital administrator

"I need a centralized scheduling system that simplifies staff management, tracks availability, and ensures compliance with minimal manual effort."

Goals

- Streamline staff scheduling to reduce manual workload and errors.
- Ensure compliance with labor regulations and hospital policies efficiently.

Frustrations

- Labor-intensive manual scheduling that is prone to errors and inefficiencies.
- Difficulty tracking staff availability and managing compliance with labor regulations.

Rajesh Verma, a 45-year-old hospital administrator, oversees staff scheduling and compliance. He struggles with labor-intensive manual processes and tracking staff availability efficiently.

User journey map



Rajesh Verma
Hospital Administrator



Journey goal- Goal: Efficiently manage her schedule, accommodate emergency changes, and integrate with her personal calendar.

 ACTION	Recognizes inefficiencies in current system	Evaluates different scheduling software	Chooses a scheduling app and seeks board approval	Oversees implementation and organizes training	Monitors system performance and gathers feedback
 TASK LIST	Tasks- 1. Identify issues with current system 2. Research automated solutions 3. Discuss with peers	Tasks- 1. Attend demos/webinars 2. Seek peer feedback 3. Compare features and pricing	Tasks- 1. Prepare presentation for board 2. Collect necessary documentation 3. Address board's concerns	Tasks- 1. Coordinate with vendor 2. Schedule training sessions 3. Monitor initial rollout	Tasks- 1. Collect staff feedback 2. Review system performance reports 3. Make necessary adjustments
 FEELING ADJECTIVE	Frustrated	Hopeful	Confident	Determined	Satisfied
 IMPROVEMENT OPPORTUNITIES	Create awareness through targeted marketing	Offer free trials and Provide detailed feature comparisons	Provide comprehensive support materials and Testimonials	Add discussion forum	Establish feedback mechanisms and Regular system reviews

miro

Competitive Audit

To conduct a competitive analysis of staff scheduling apps, we will evaluate 3 apps in the market.

Here's an overview of their features and functionalities:



schedulicity

Schedulicity offers scheduling tools, automated reminders, and online booking capabilities. It also provides client management features, such as customer profiles and payment processing.

Pros -

- Intuitive and clean interface
- Robust scheduling capabilities
- Integrated payment processing
- marketing tools for promoting events or specials

Cons -

- Limited team communication features
- Does not have specialized features for managing shift rotations or specific requirements



deputy

Deputy offers scheduling tools, automated reminders, and online booking capabilities. It also provides client management features, such as customer profiles and payment processing.

Pros -

- Comprehensive scheduling and time management features
- Integration with payroll systems
- shift swapping and availability management
- Real time communication with employees

Cons -

- Steeper learning curve for administrators
- Some advanced features may require higher subscription tiers



When I Work

When I work focuses on easy scheduling, time tracking and team communication. It offers shift templates, automated scheduling, and built-in messaging capabilities.

Pros -

- Simple and intuitive interface
- Automated scheduling based on availability and preferences
- Shift swapping and time-off management
- Mobile app for on-the-go access

Cons -

- limited reporting and analytics features
- may lack more advanced scheduling capabilities required for complex operations

miro

Insights

After analyzing and over-viewing these competitors. I came to the following conclusions for my app:

- Simplify the process of creating and managing staff schedules for administrators.
- Keep track of unexpected changes in the schedules.
- Incorporate a notification system to inform staff members about their assigned shifts and any schedule changes.
- Cater to customisation needs.
- Simple UI that isnt heavy on mobile devices.

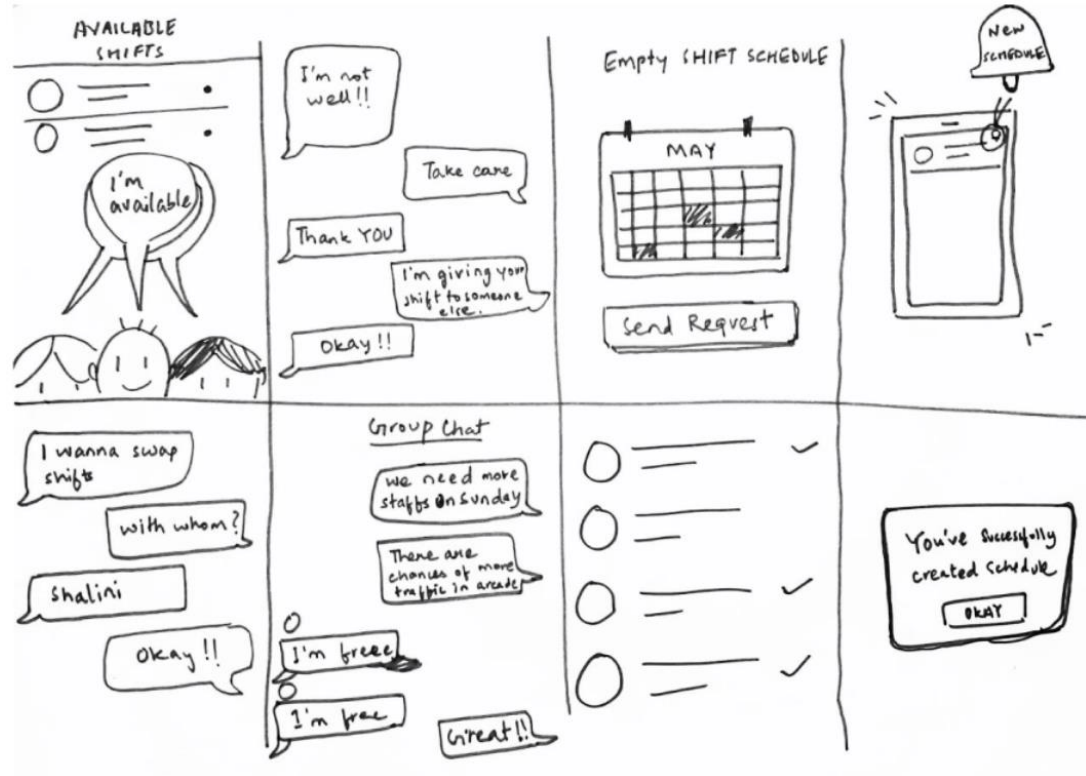
How might we ?

'How might we' queries serve as starting points for ideation and brainstorming sessions , encouraging creative thinking and generating potential solutions .

1. How might we simplify the process of creating and managing staff schedules for the hospital ?
2. How might we keep track of unexpected changes in the schedules ?
3. How might we incorporate a notification system to inform staff members about their assigned shifts and any schedule changes ?
4. How might we offer a comprehensive overview of staff availability and scheduling conflicts to help managers make an informed decision ?

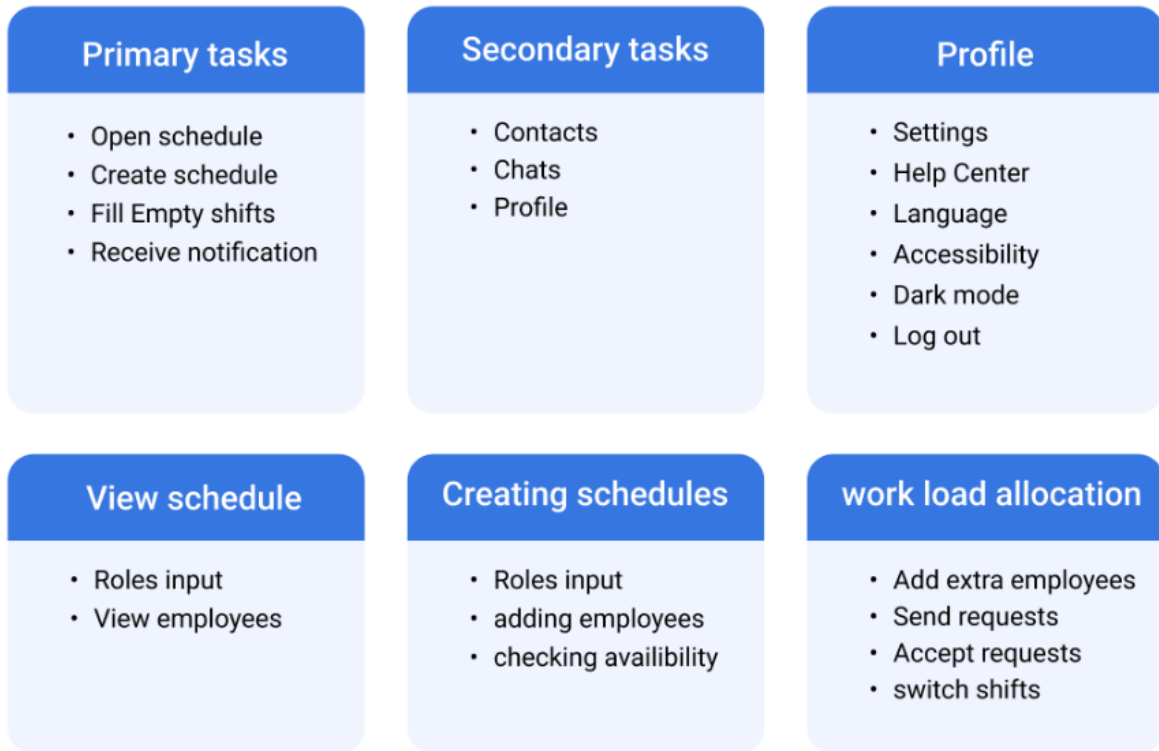
Crazy Eight

I utilised the Crazy eight method, a rapid ideation exercise within the design thinking framework, to sketch eight distinct ideas in just eight minutes. This exercise allowed me to generate a diverse range of ideas that cater to the needs of both users.



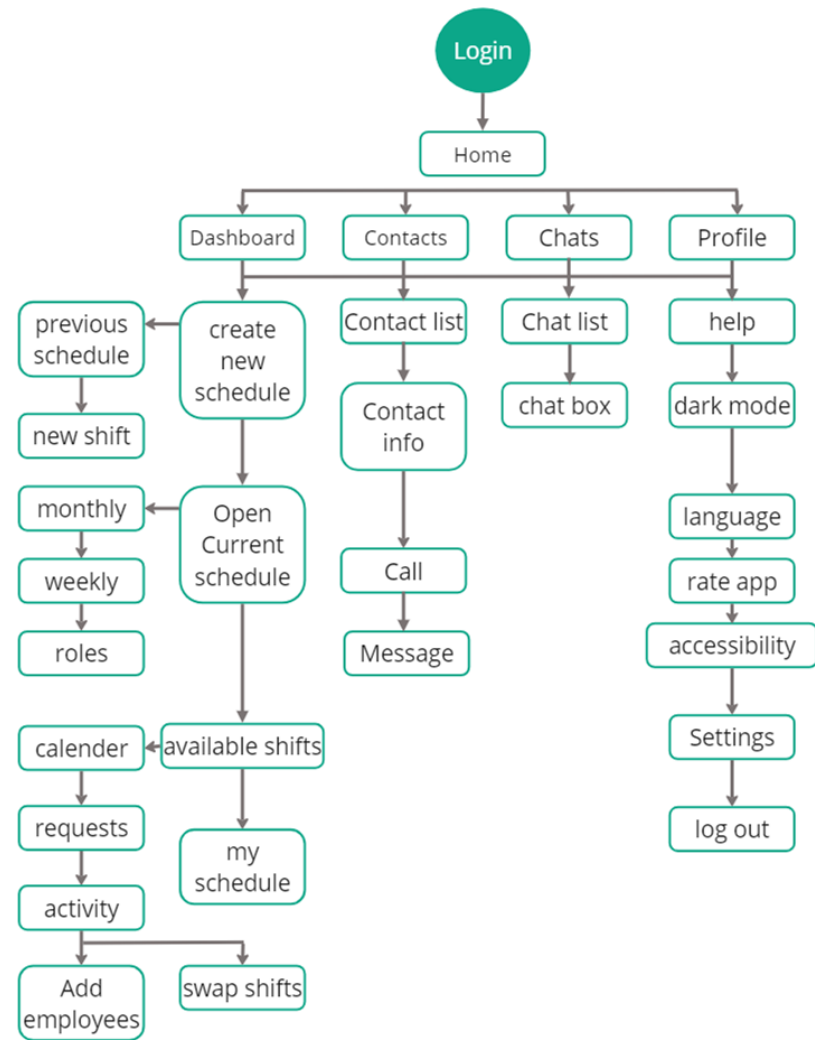
Card Sorting

This activity involves organising a set of features and functionalities into categories based on their perceived relevance or similarity. It helps to understand how users group and prioritise different elements of the app.



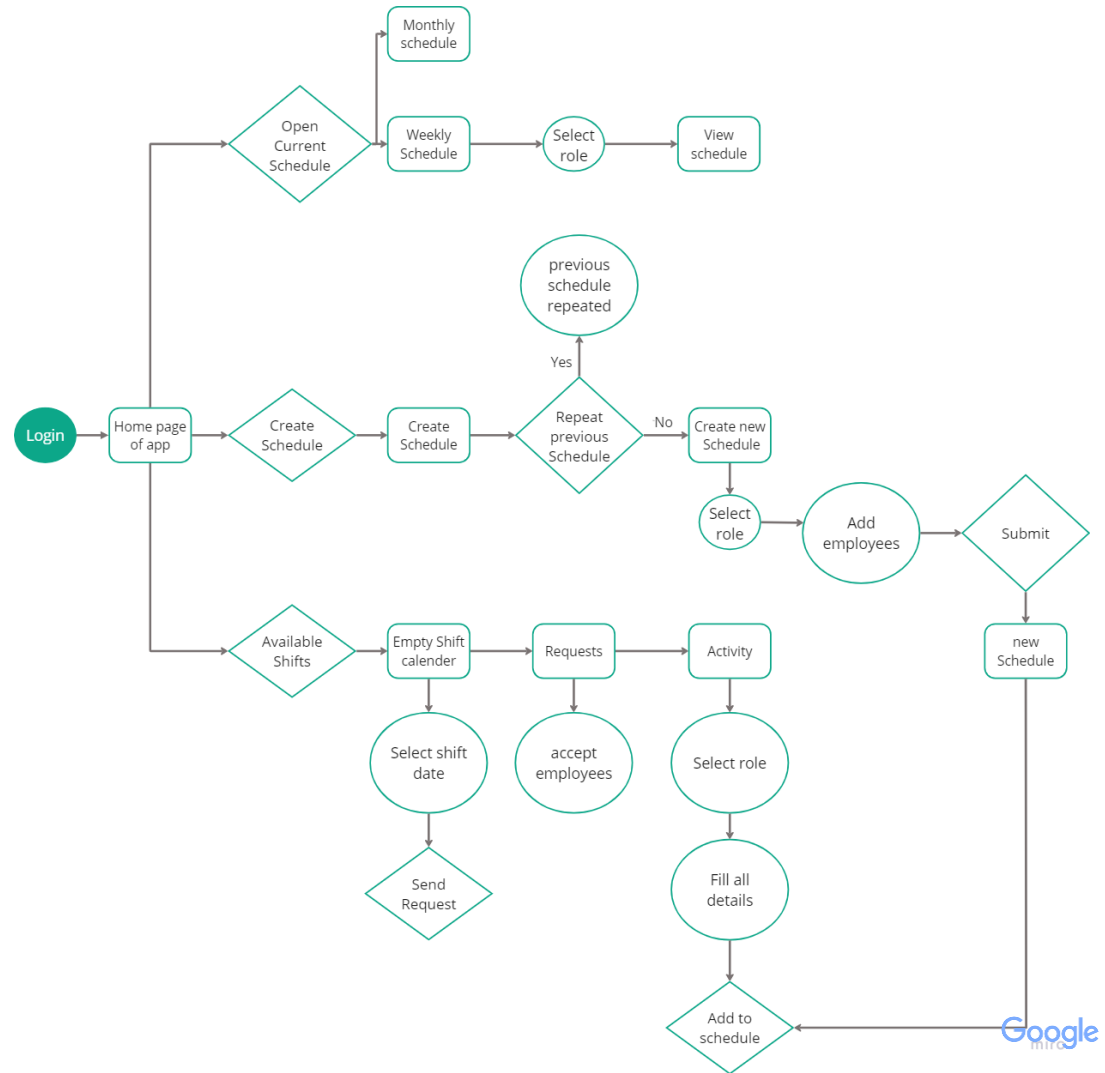
Information Architecture

Information architecture (IA) focuses on organizing, structuring, and labeling content in an effective and sustainable way.



User Flow

After the defining process, I created a user flow to understand how it could be structured for the user to interact with the app. User flow allowed me to see how the user could complete a task from start to finish.



Storyboard

As part of the exploration into the users experience with the staff scheduling app, I created two storyboards. The big picture storyboard dived into the users interaction with the app and its overall uselessness, while the close-up storyboard focused on the users actions during screen transitions.

UX Design Storyboard

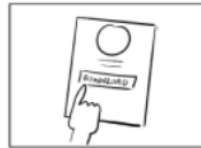
Scenario: Use the App to manage the staff schedule — big picture.



Receives a call for leave.



Employees are swapping shifts without informing beforehand.



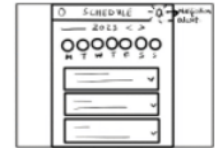
Suman downloads the staff scheduling app.



Suman opens the app.



Suman opens the weekly schedule.



Suman receives a notification.



Suman asks the employee to download the same app.



Suman receives leave notification in the app and makes changes.



Suman is happy with entering times if no Email is required.



An employee requests for leave.



Suman chats with her employee in the chatbox.



Suman finds a replacement for the shift and adds them to the schedule.

Usability study: findings

This study evaluates the new hospital staff scheduling app's effectiveness and user satisfaction. It involved doctors, nurses, and administrators, focusing on intuitiveness, flexibility, and workflow efficiency improvements.

Round 1 findings

- 1 Users found the new system more intuitive and time-saving.
- 2 Quick shift swaps and leave requests improved flexibility.
- 3 Real-time notifications enhanced communication and reduced stress.

Round 2 findings

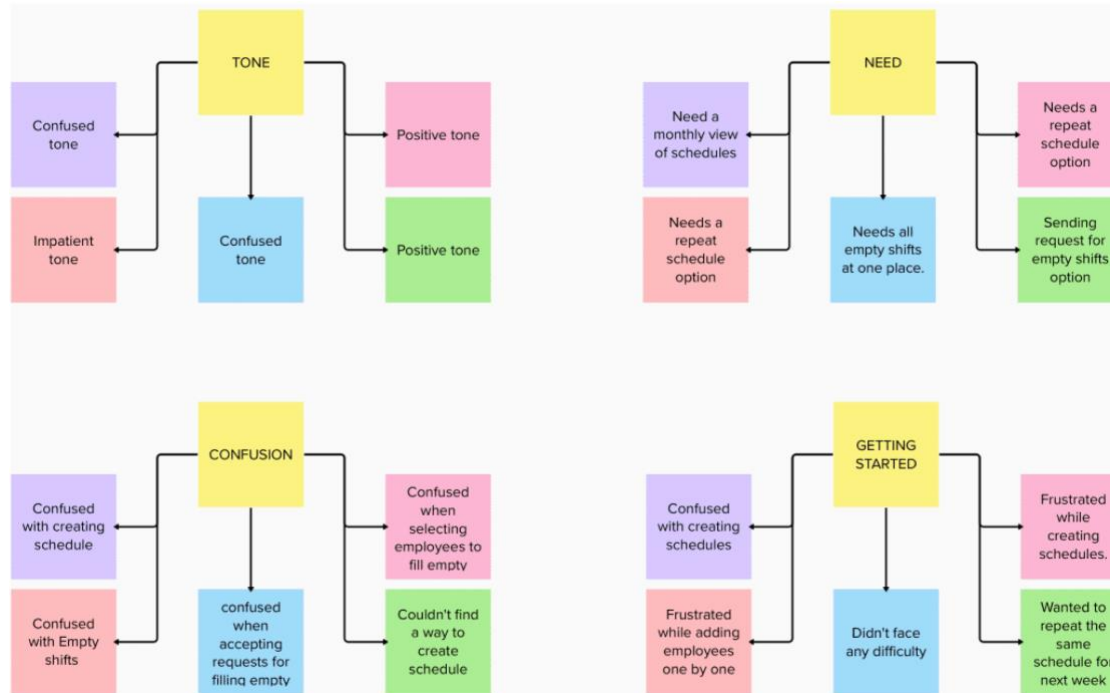
- 1 Accessing schedules from any device was highly valued.
- 2 Automated features minimized manual errors.
- 3 Integration with personal calendars improved work-life balance.

Refining the design

- Mockups
- High-fidelity prototype
- Accessibility

Affinity Mapping

Once finishing the usability testing session for the lo-fi prototype, I conducted the affinity map to arrange their feedback into a few groups. It helped me find the common patterns in these observations.

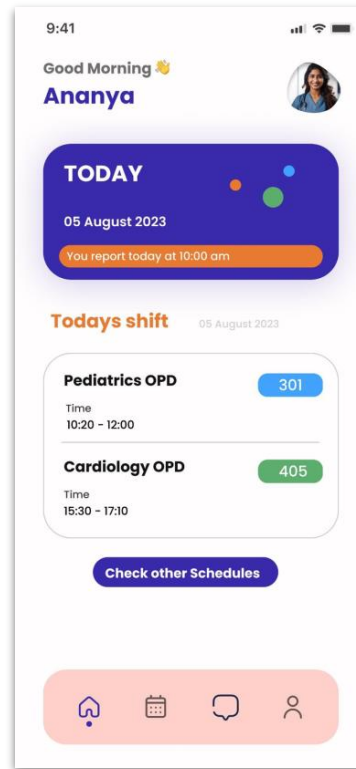


Mockups

Before usability study



After usability study

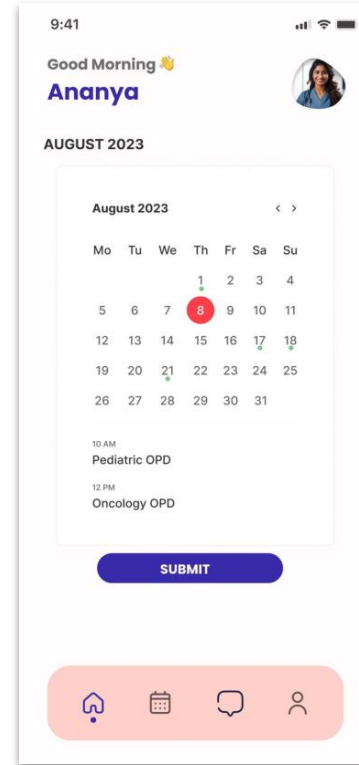


Mockups

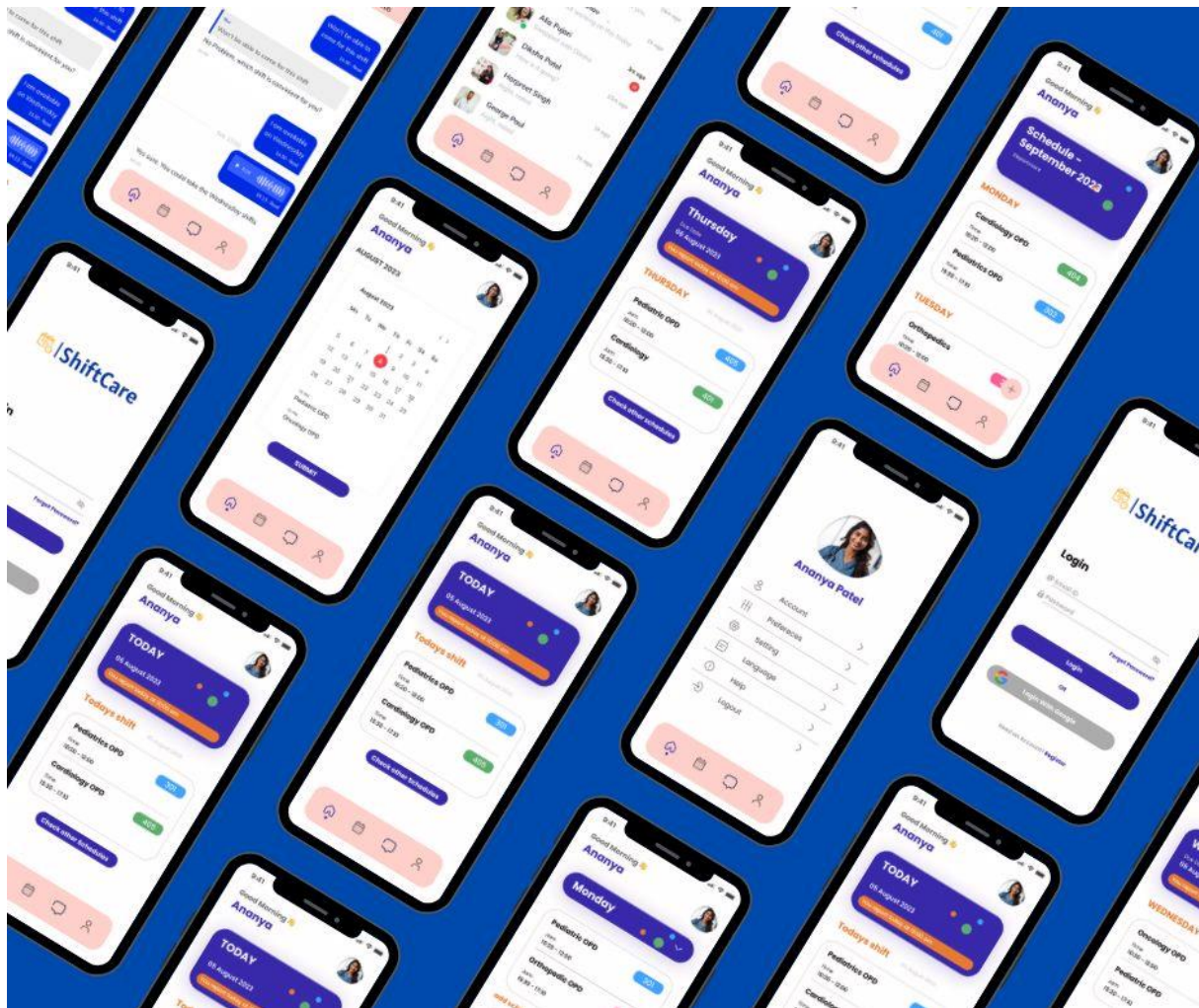
Before usability study



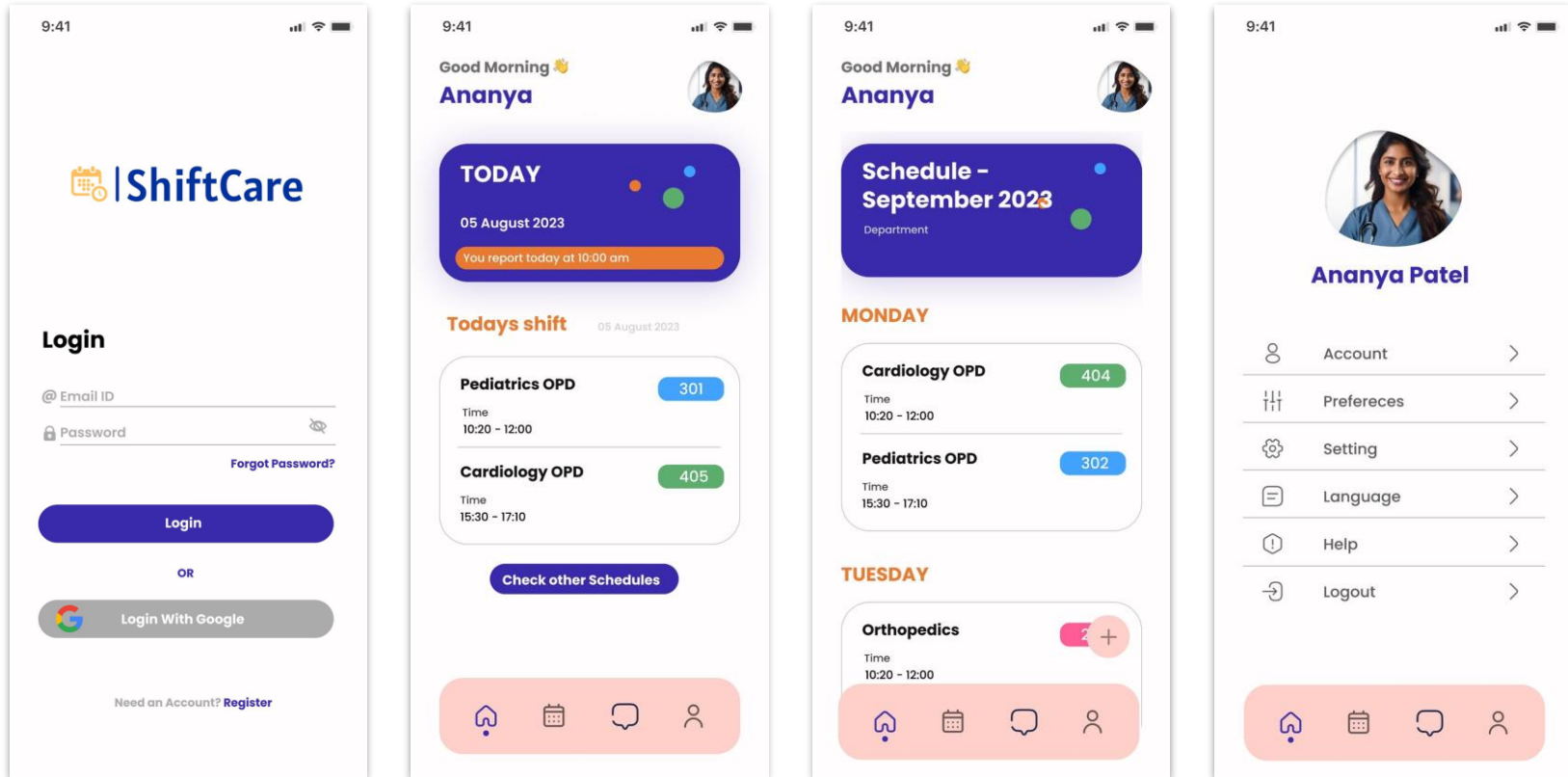
After usability study



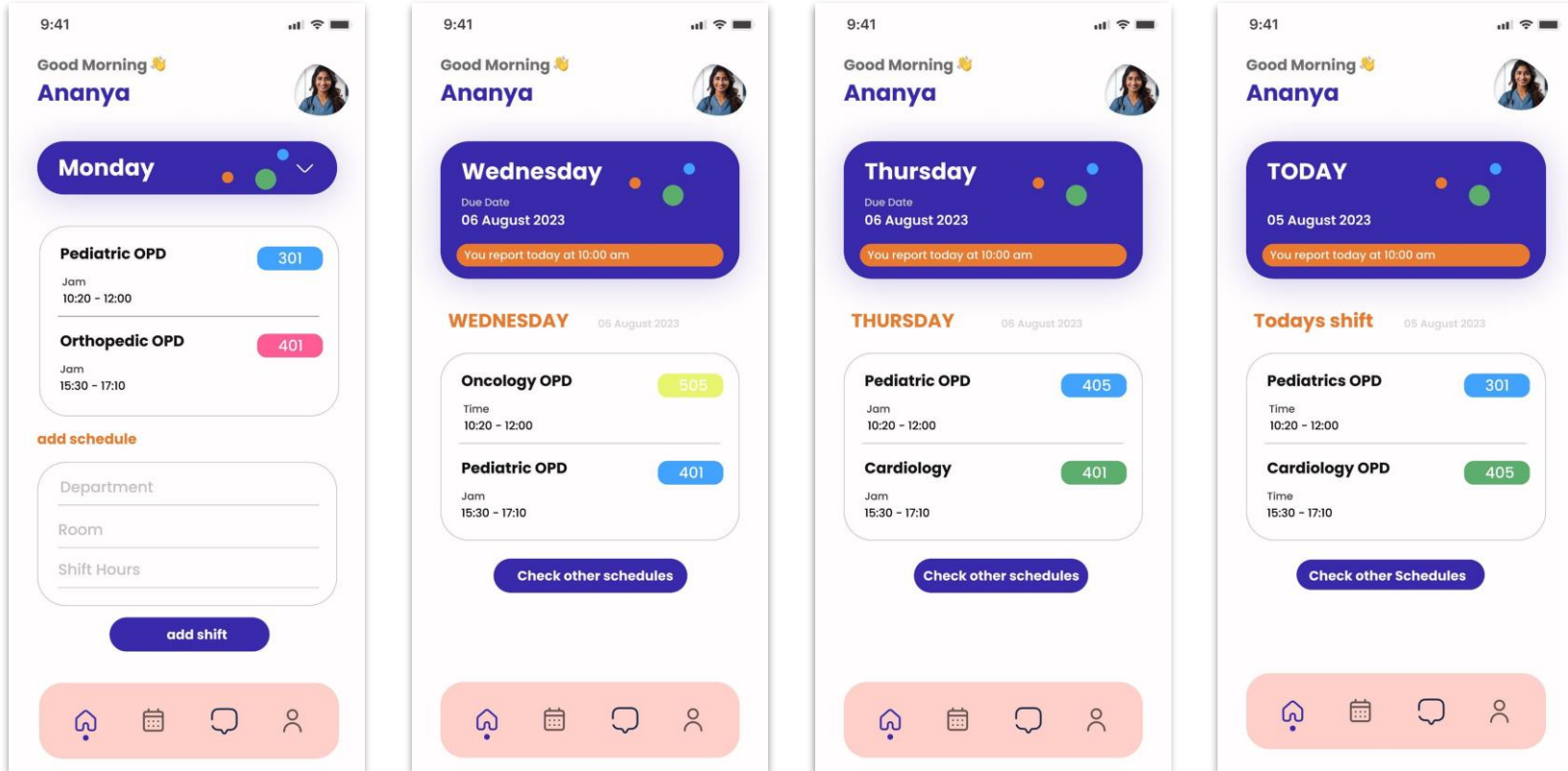
Mockups



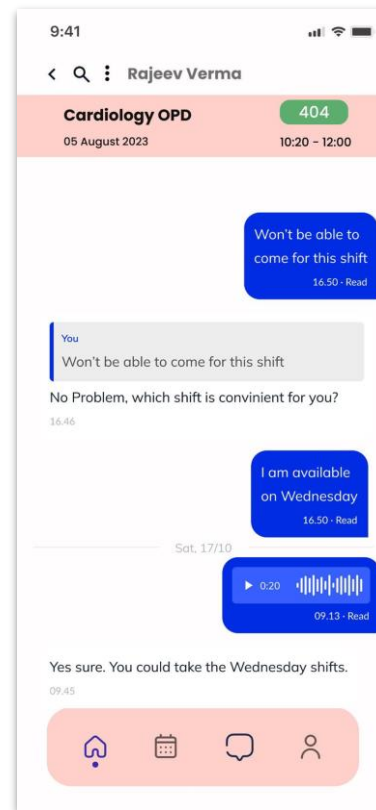
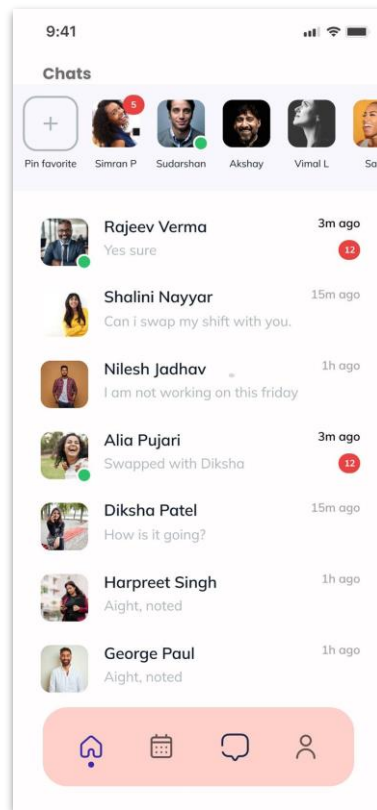
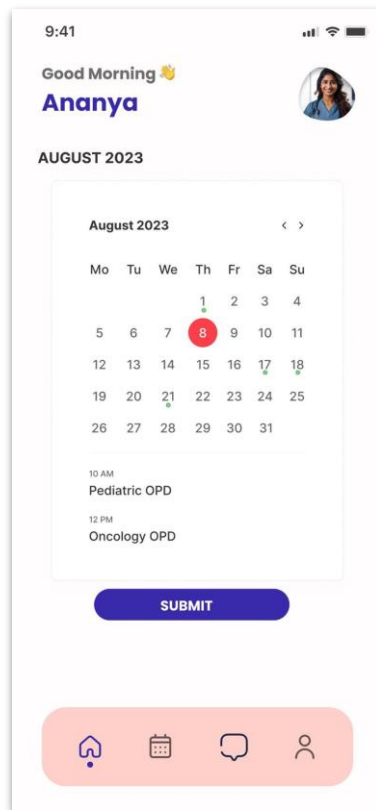
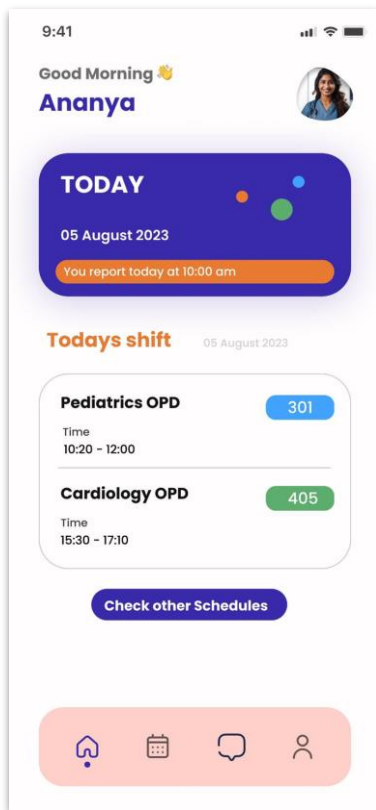
High-fidelity prototype



High-fidelity prototype



High-fidelity prototype



Accessibility considerations

1

Color Contrast

High color contrast ratios were used between text and backgrounds to ensure readability for users with visual impairments.

2

Keyboard Navigation

All interactive elements are accessible via keyboard navigation, facilitating use for individuals with mobility impairments.

3

Screen Reader Compatibility

ARIA landmarks and labels were implemented to ensure screen readers can accurately interpret the app's content for visually impaired users.

Going forward

- Takeaways
- Next steps

Takeaways



Impact:

The design has made scheduling shifts simpler and more intuitive, greatly enhancing the user experience. A study participant shared, “I no longer dread checking my schedule. It’s clear, easy to navigate, and has made my work life so much smoother.”



What I learned:

I learned the value of understanding user needs and the impact of accessibility in design. Collaborating with peers highlighted the importance of feedback in refining user experiences. This project reinforced how user-centric design can effectively address real-world challenges.

1

Conduct User Testing

Perform additional user testing to gather more feedback and refine the app based on real-world usage.

2

Implement Advanced Analytics

Integrate analytics tools to track user interactions and identify areas for improvement.

3

Optimize for Mobile

Expand the design to ensure compatibility and usability across mobile devices, enhancing accessibility and reach.

Let's connect!



Feel free to reach out for any inquiries or to review more of my work. You can contact me at ankitavthatte@gmail.com or visit my portfolio at <https://www.behance.net/ankitathatte>.